

**ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN
KHOA KHOA HỌC VÀ KỸ THUẬT THÔNG TIN**

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**BÁO CÁO THỰC HÀNH 1
MÔN HỌC: CÔNG NGHỆ DỮ LIỆU LỚN
(IE212.Q21)**

Sinh viên thực hiện: Lê Nguyễn Thùy Dương - 22520298
Giảng viên hướng dẫn: Nguyễn Hiếu Nghĩa

Thành phố Hồ Chí Minh, tháng 3 năm 2026

Bài 1: Tính Điểm Đánh Giá Trung Bình và Tổng Số Lượt Đánh Giá Cho Mỗi Phim

Ảnh chụp màn hình kết quả:

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -cat /output_bai1/part-r-00000
Toy Story (1995) Average rating: 4.25 (Total ratings: 2)
The Shawshank Redemption (1994) Average rating: 4.00 (Total ratings: 2)
Back to the Future (1985) Average rating: 4.00 (Total ratings: 2)
Gladiator (2000) Average rating: 3.25 (Total ratings: 2)
Braveheart (1995) Average rating: 4.25 (Total ratings: 2)
Jurassic Park (1993) Average rating: 4.00 (Total ratings: 2)
The Departed (2006) Average rating: 3.25 (Total ratings: 2)
The Silence of the Lambs (1991) Average rating: 3.50 (Total ratings: 2)
Saving Private Ryan (1998) Average rating: 4.00 (Total ratings: 2)
Avatar (2009) Average rating: 4.75 (Total ratings: 2)
Interstellar (2014) Average rating: 3.50 (Total ratings: 2)
The Prestige (2006) Average rating: 4.00 (Total ratings: 2)
The Avengers (2012) Average rating: 3.75 (Total ratings: 2)
Memento (2000) Average rating: 4.00 (Total ratings: 2)
The Green Mile (1999) Average rating: 3.50 (Total ratings: 2)
The Matrix (1999) Average rating: 3.75 (Total ratings: 2)
The Lord of the Rings: The Fellowship of the Ring (2001) Average rating: 4.25 (Total ratings: 2)
The Lord of the Rings: The Two Towers (2002) Average rating: 3.75 (Total ratings: 2)
The Lord of the Rings: The Return of the King (2003) Average rating: 3.50 (Total ratings: 2)
Fight Club (1999) Average rating: 4.25 (Total ratings: 2)
Se7en (1995) Average rating: 3.75 (Total ratings: 2)
The Usual Suspects (1995) Average rating: 4.25 (Total ratings: 2)
Schindler's List (1993) Average rating: 4.25 (Total ratings: 2)
American Beauty (1999) Average rating: 3.75 (Total ratings: 2)
Good Will Hunting (1997) Average rating: 3.50 (Total ratings: 2)
The Social Network (2010) Average rating: 3.75 (Total ratings: 2)
Slumdog Millionaire (2008) Average rating: 4.25 (Total ratings: 2)
Whiplash (2014) Average rating: 3.75 (Total ratings: 2)
La La Land (2016) Average rating: 4.00 (Total ratings: 2)
Inception (2010) Average rating: 3.75 (Total ratings: 2)
Mad Max: Fury Road (2015) Average rating: 4.00 (Total ratings: 2)
The Revenant (2015) Average rating: 3.50 (Total ratings: 2)
The Dark Knight (2008) Average rating: 3.50 (Total ratings: 2)
Gravity (2013) Average rating: 4.25 (Total ratings: 2)
The Wolf of Wall Street (2013) Average rating: 4.25 (Total ratings: 2)
Birdman (2014) Average rating: 3.00 (Total ratings: 2)
Pulp Fiction (1994) Average rating: 4.00 (Total ratings: 2)
The Big Short (2015) Average rating: 4.50 (Total ratings: 2)
Manchester by the Sea (2016) Average rating: 4.00 (Total ratings: 2)
Moonlight (2016) Average rating: 3.50 (Total ratings: 2)
Spotlight (2015) Average rating: 4.25 (Total ratings: 2)
The Imitation Game (2014) Average rating: 3.75 (Total ratings: 2)
Forrest Gump (1994) Average rating: 3.75 (Total ratings: 2)

Dunkirk (2017) Average rating: 3.50 (Total ratings: 2)
Coco (2017) Average rating: 4.25 (Total ratings: 2)
Parasite (2019) Average rating: 4.00 (Total ratings: 2)
The Godfather (1972) Average rating: 3.50 (Total ratings: 2)
Titanic (1997) Average rating: 3.50 (Total ratings: 2)
Star Wars: A New Hope (1977) Average rating: 3.50 (Total ratings: 2)
The Lion King (1994) Average rating: 4.00 (Total ratings: 2)
```

Các file kết quả đầu ra.

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -ls /output_bai1
Found 2 items
-rw-r--r-- 1 duong supergroup 0 2026-03-22 00:38 /output_bai1/_SUCCESS
-rw-r--r-- 1 duong supergroup 3136 2026-03-22 00:38 /output_bai1/part-r-00000
```

Bài 2: Phân Tích Đánh Giá Theo Thể Loại

Ảnh chụp màn hình kết quả:

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -cat /output_bai2/part-r-00000
Action Avg: 3.73, Count: 20
Adventure Avg: 3.85, Count: 30
Animation Avg: 4.17, Count: 6
Biography Avg: 4.14, Count: 14
Children Avg: 4.25, Count: 2
Comedy Avg: 3.96, Count: 12
Crime Avg: 3.78, Count: 20
Drama Avg: 3.81, Count: 76
Family Avg: 4.25, Count: 2
Fantasy Avg: 3.95, Count: 10
History Avg: 4.00, Count: 6
Music Avg: 3.88, Count: 4
Mystery Avg: 4.00, Count: 8
Romance Avg: 3.75, Count: 8
Sci-Fi Avg: 3.85, Count: 20
Thriller Avg: 3.88, Count: 16
War Avg: 4.00, Count: 2
```

Các file kết quả đầu ra.

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -ls /output_bai2
Found 2 items
-rw-r--r-- 1 duong supergroup 0 2026-03-22 01:01 /output_bai2/_SUCCESS
-rw-r--r-- 1 duong supergroup 479 2026-03-22 01:01 /output_bai2/part-r-00000
```

Bài 3: Phân Tích Đánh Giá Theo Giới Tính

Ảnh chụp màn hình kết quả:

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -cat /output_bai3/part-r-00000
Toy Story (1995)      Male: 4.50, Female: 4.00
The Shawshank Redemption (1994) Male: 3.00, Female: 5.00
Back to the Future (1985)      Male: 4.00, Female: 4.00
Gladiator (2000)      Male: 3.50, Female: 3.00
Braveheart (1995)      Male: 4.50, Female: 4.00
Jurassic Park (1993)      Male: 3.50, Female: 4.50
The Departed (2006)      Male: 4.50, Female: 2.00
The Silence of the Lambs (1991) Male: 3.50, Female: 3.50
Saving Private Ryan (1998)      Male: 4.00, Female: 4.00
Avatar (2009)      Male: 5.00, Female: 4.50
Interstellar (2014)      Male: 3.50, Female: 3.50
The Prestige (2006)      Male: 4.00, Female: 4.00
The Avengers (2012)      Male: 3.50, Female: 4.00
Memento (2000)      Male: 3.50, Female: 4.50
The Green Mile (1999)      Male: 3.00, Female: 4.00
The Matrix (1999)      Male: 3.00, Female: 4.50
The Lord of the Rings: The Fellowship of the Ring (2001)      Male: 4.50, Female: 4.00
The Lord of the Rings: The Two Towers (2002)      Male: 4.50, Female: 3.00
The Lord of the Rings: The Return of the King (2003)      Male: 4.00, Female: 3.00
Fight Club (1999)      Male: 5.00, Female: 3.50
Se7en (1995)      Male: 4.00, Female: 3.50
The Usual Suspects (1995)      Male: 4.00, Female: 4.50
Schindler's List (1993) Male: 4.00, Female: 4.50
American Beauty (1999)      Male: 3.00, Female: 4.50
Good Will Hunting (1997)      Male: 3.00, Female: 4.00
The Social Network (2010)      Male: 3.50, Female: 4.00
Slumdog Millionaire (2008)      Male: 4.50, Female: 4.00
Whiplash (2014) Male: 4.50, Female: 3.00
La La Land (2016)      Male: 5.00, Female: 3.00
Inception (2010)      Male: 3.50, Female: 4.00
Mad Max: Fury Road (2015)      Male: 4.00, Female: 4.00
The Revenant (2015)      Male: 3.50, Female: 3.50
The Dark Knight (2008) Male: 4.00, Female: 3.00
Gravity (2013)      Male: 4.00, Female: 4.50
The Wolf of Wall Street (2013) Male: 4.00, Female: 4.50
Birdman (2014)      Male: 3.00, Female: 3.00
Pulp Fiction (1994)      Male: 4.50, Female: 3.50
The Big Short (2015)      Male: 5.00, Female: 4.00
Manchester by the Sea (2016)      Male: 3.50, Female: 4.50
Moonlight (2016)      Male: 3.00, Female: 4.00
Spotlight (2015)      Male: 4.50, Female: 4.00

The Imitation Game (2014)      Male: 4.50, Female: 3.00
Forrest Gump (1994)      Male: 2.50, Female: 5.00
Dunkirk (2017)      Male: 4.00, Female: 3.00
Coco (2017)      Male: 5.00, Female: 3.50
Parasite (2019) Male: 4.00, Female: 4.00
The Godfather (1972)      Male: 4.50, Female: 2.50
Titanic (1997)      Male: 4.00, Female: 3.00
Star Wars: A New Hope (1977)      Male: 4.00, Female: 3.00
The Lion King (1994)      Male: 4.50, Female: 3.50
```

Các file kết quả đầu ra.

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -ls /output_bai3
Found 2 items
-rw-r--r--  1 duong supergroup      0 2026-03-22 01:12 /output_bai3/_SUCCESS
-rw-r--r--  1 duong supergroup 2386 2026-03-22 01:12 /output_bai3/part-r-00000
```

Bài 4: Phân Tích Đánh Giá Theo Nhóm Tuổi

Ảnh chụp màn hình kết quả:

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -cat /output_bai4/part-r-00000
Toy Story (1995)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
The Shawshank Redemption (1994) 0-18: NA 18-35: 4.00 35-50: NA 50+: NA
Back to the Future (1985)      0-18: NA 18-35: 4.00 35-50: NA 50+: NA
Gladiator (2000)      0-18: NA 18-35: 3.00 35-50: 3.50 50+: NA
Braveheart (1995)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Jurassic Park (1993)      0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Departed (2006)      0-18: NA 18-35: 2.00 35-50: 4.50 50+: NA
The Silence of the Lambs (1991) 0-18: NA 18-35: 3.50 35-50: 3.50 50+: NA
Saving Private Ryan (1998)      0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
Avatar (2009)      0-18: NA 18-35: 4.50 35-50: 5.00 50+: NA
Interstellar (2014)      0-18: NA 18-35: 3.50 35-50: 3.50 50+: NA
The Prestige (2006)      0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
The Avengers (2012)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Memento (2000)      0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
The Green Mile (1999)      0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
The Matrix (1999)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
The Lord of the Rings: The Two Towers (2002)      0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
The Lord of the Rings: The Return of the King (2003)      0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Fight Club (1999)      0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Se7en (1995)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Usual Suspects (1995)      0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Schindler's List (1993)      0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
American Beauty (1999)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Good Will Hunting (1997)      0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Social Network (2010)      0-18: NA 18-35: 4.00 35-50: 3.50 50+: NA
Slumdog Millionaire (2008)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Whiplash (2014)      0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
La La Land (2016)      0-18: NA 18-35: 3.00 35-50: 5.00 50+: NA
Inception (2010)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Mad Max: Fury Road (2015)      0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Revenant (2015)      0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Dark Knight (2008)      0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
Gravity (2013)      0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
The Wolf of Wall Street (2013)      0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
Birdman (2014)      0-18: NA 18-35: 3.00 35-50: NA 50+: NA
Pulp Fiction (1994)      0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Big Short (2015)      0-18: NA 18-35: 4.50 35-50: NA 50+: NA
Manchester by the Sea (2016)      0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
Moonlight (2016)      0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
Spotlight (2015)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
The Imitation Game (2014)      0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
Forrest Gump (1994)      0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Dunkirk (2017)      0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Coco (2017)      0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Parasite (2019)      0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
The Godfather (1972)      0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Titanic (1997)      0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
Star Wars: A New Hope (1977)      0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
The Lion King (1994)      0-18: NA 18-35: 4.00 35-50: NA 50+: NA
```

Các file kết quả đầu ra.

```
duong@ubuntu:/mnt/hgfs/LAB1$ hdfs dfs -ls /output_bai4
Found 2 items
-rw-r--r--  1 duong supergroup      0 2026-03-22 01:23 /output_bai4/_SUCCESS
-rw-r--r--  1 duong supergroup 3142 2026-03-22 01:23 /output_bai4/part-r-00000
```

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